Introduction to Numerical Methods 11

Boundary value problems

Problem to solve: $\frac{dx}{dt} = f(x,t)$, $x(t_0) = x_0$ $x(t) = ?$

Discrete approximation

$dt \to \Delta t$, $dx \to \Delta x$ $\qquad t_n = t_0 + n\Delta t$

Euler method

$$x_{n+1} = x_n + f(x_n, t_n)\,\Delta t$$

Stability of the Euler method

e.g. $f(x,t) = -\xi x$

$$x_{n+1} = (1 + \xi\Delta t)\,x_n$$

converges if $|1 + \xi\Delta t| < 1$

Any dimensions: $\frac{d\underline{x}}{dt} = \underline{f}(\underline{x}, t)$

↳ eigensystem: the most important is the largest eigenvalue: λ

Euler method converges if $|1 + \lambda\Delta t| < 1$

Implicit Euler

$$x_{n+1} = x_n + f(x_{n+1}, t_{n+1})\,\Delta t$$

we need to solve it for x_{n+1}

in case of $f(x,t) = -\xi x$

Stability of the Implicit Euler for $f(x,t) = -k x$

$$x_{n+1} = \frac{x_n}{1+k\Delta t}$$

which is stable for $k > 0 \Rightarrow$ For all stable systems it is stable !!

Error $\mathcal{O}(h)$

Higher order methods

$(h = \Delta t)$

Runge - Kutta method

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$t_{n+1} = t_n + h$$

$k_1 = f(x_n, t_n)$	Slope at the beg.
$k_2 = f(x_n + h\frac{k_1}{2}, t_n + \frac{h}{2})$	Slope at midpoint
$k_3 = f(x_n + h\frac{k_2}{2}, t_n + \frac{h}{2})$	Slope at better midpoint
$k_4 = f(x_n + h k_3, t_n + h)$	Slope at the end

Error: $\mathcal{O}(h^5)$

RK4 vs.	Implicit Euler
- more precise $\mathcal{O}(h^5)$ vs $\mathcal{O}(h)$	- always stable
- use this unless issues	- less calculations

with stability or function evaluation

In many applications $f(x, \cancel{t})$

Second order differential equation

(Newton's equation of motion)

$$\ddot{x} = \frac{F}{m} = f(x,t) \quad , \quad x(t=0) = x_0 \, , \; v(t=0) = v_0$$

Euler method

$$v_{n+1} = v_n + f(x_n)\,\Delta t$$
$$x_{n+1} = x_n + v_n\,\Delta t$$

Implicit Euler method

for position use new velocity

$$v_{n+1} = v_n + f(x_n)\,\Delta t$$
$$x_{n+1} = x_n + v_{\boxed{n+1}}\,\Delta t$$

Symplectic integrator: conserves the structure of the equations, keeps energy (including some virtual terms) constant

The Euler method is not symplectic

The implicit Euler is a symplectic integrator

Higher order methods

Runge-Kutta works but is not a symplectic integrator.

General standard: Velocity Verlet

$$x(t+\Delta t) = x(t) + v(t)\Delta t + \frac{1}{2} a(t) \Delta t^2$$

$$v(t+\Delta t) = v(t) + \frac{a(t) + a(t+\Delta t)}{2} \Delta t$$

Algorithm:

1. $x(t+\Delta t) = x(t) + v(t)\Delta t + \frac{1}{2} a(t)\Delta t^2$
2. get $a(t+\Delta t)$ using the interactions at $x(t+\Delta t)$
3. $v(t+\Delta t) = v(t) + \frac{1}{2}\left(a(t) + a(t+\Delta t)\right)\Delta t$

Also a symplectic integrator